BIG DATA ANALYTICS — CA3 ACTIVITY 2

Bengaluru Urban Mobility & Transit Analytics

Multi-modal (Bike · Auto · Cab) demand, congestion, fare and ETA analytics with MapReduce and machine learning

247,951	4	2	10
Trips analysed	MapReduce jobs	ML models	City hubs

TEAM MEMBERS

Rupesh Singh USN: 25MCAR0017

Anoon MR USN: 25MCAR0179

Pankaj Singh USN: 25MCAR0223

MCA AIML — III SEMESTER

Dhanya Varsha USN: 25MCAR0201

Chiranth PG USN: 25MCAR0169

WHY THIS PROJECT

The real-world problem

Bengaluru riders choose between bike taxis, autos and cabs on roads that slow to a crawl at peak. Riders, drivers and city planners each need different answers from the same trip data.

1 When and where is demand highest?

By hour, hub and vehicle type.

2 Which ride should I take?

ETA and fare for bike vs auto vs cab.

3 Where are the choke points?

Slowest peak-hour corridors.

4 How do peaks and rain change trips?

Speed, delay and surge.

5 Where should drivers wait?

Fare earned per minute, by hour and corridor.

6 Can ML predict ETA and demand?

Regression for ETA, classification for surge.

STAGE 0

Dataset: synthetic Bengaluru trips

- No open trip-level dataset exists for Bengaluru bike / auto / cab rides, so trips are generated from documented city rules (seed 42, fully reproducible).
- 10 hubs with real coordinates, from Koramangala and Whitefield to Kempegowda Airport and Majestic.
- Traffic peaks 08:30–11:30 (1.5–2.2×) and 17:30–21:30 (1.6–2.4×); free-flow at night; rain adds 15%.
- Rate cards: Bike ₹20 + ₹8/km, Auto ₹30 + ₹15/km, Cab ₹60 + ₹18–22/km, plus delay charge, night surcharge and surge.
- 1 Jan – 30 Jun 2025 (181 days, 25 rainy). About 1.5% dirty rows were injected on purpose for the cleaning stage.

250,750
Raw rows

247,951
After cleaning

11
Raw columns

30
Columns after feature engineering

HOW IT ALL FITS TOGETHER

End-to-end pipeline



`pipeline/run_all.py` runs steps 0–4 in about 10 seconds. **`backend/app.py`** serves /api/dashboard?role=, /api/predict/trip and /api/predict/demand.

STAGE 2 — HADOOP-STREAMING COMPATIBLE

MapReduce implementation

Job 1 · Modal demand by hour

Map (hour | vehicle) → 1

Reduce sum → 72 rows

Job 2 · Duration & fare by zone

Map (zone | vehicle) → minutes, fare

Reduce count, averages, total → 30 rows

Job 3 · Congestion corridors

Map peak, inter-zone only: (pickup | drop) → km, min, delay

Reduce speed, delay share, status → 90 rows

Job 4 · Demand dataset

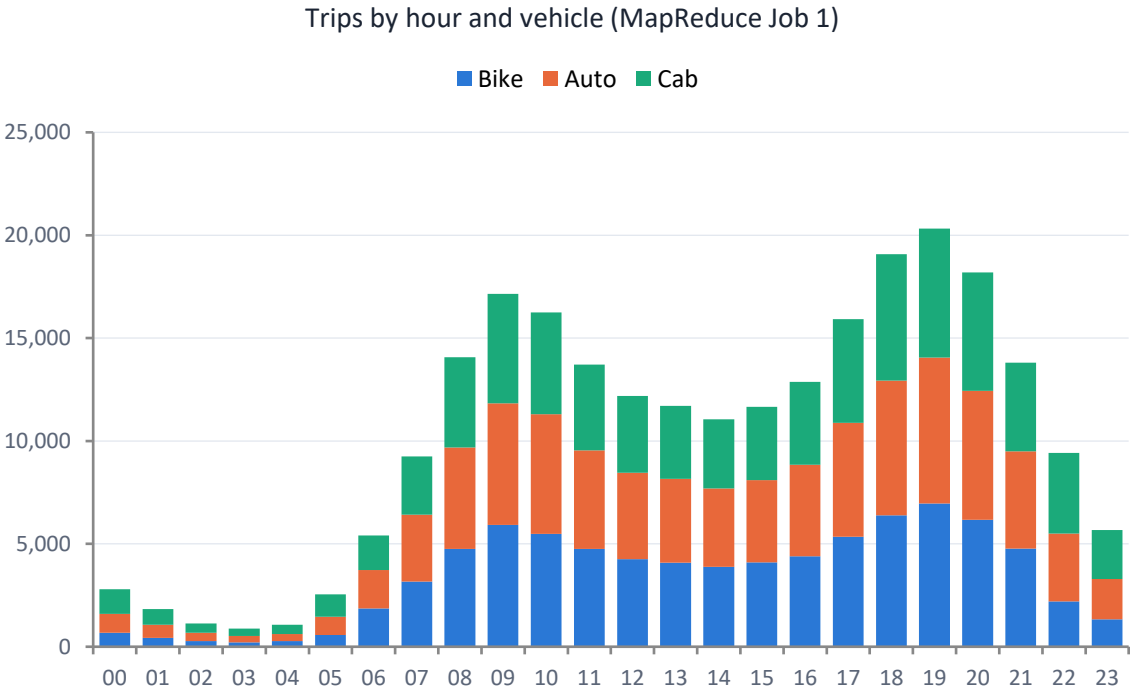
Map (date | hour | day | zone | rain) → 1

Reduce count → 38,158 rows → feeds Model B

Each job is a real mapper.py / reducer.py pair run as Map → Shuffle/Sort → Reduce subprocesses, the contract Hadoop Streaming uses. 247,951 records through all 4 jobs in 1.5 s locally.

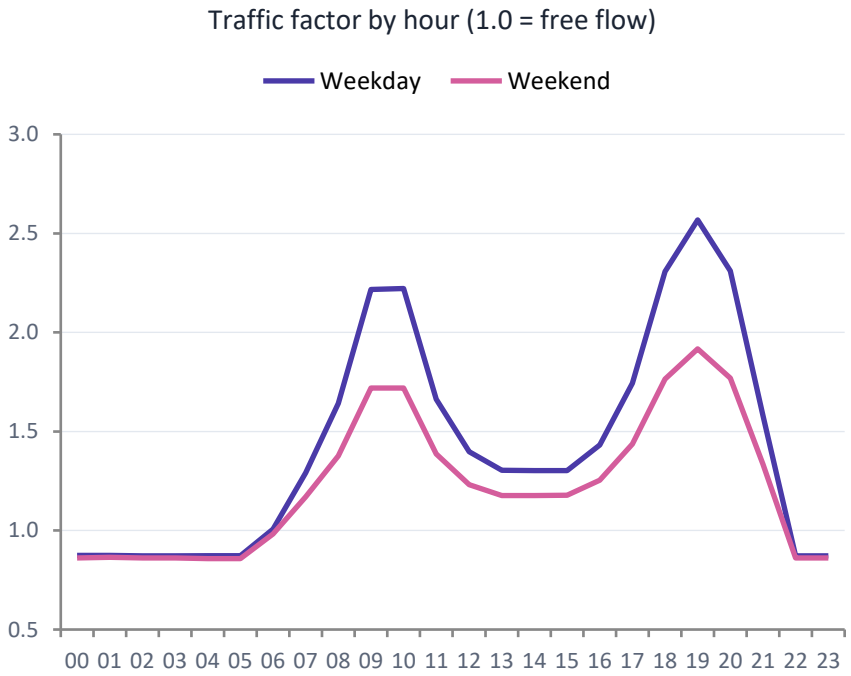
STAGE 3 — ANALYTICS

Demand peaks at 09:00 and 19:00



- Busiest hour 19:00, about 20,000 trips over the period.
- 48% of trips fall in the two peak windows.
- Mode share is balanced: bike 33%, auto 35%, cab 32%.
- Weekdays carry 76% of trips; Wednesday is busiest.

Choke points: two corridor clusters



Slowest peak-hour corridors (MapReduce Job 3)

Corridor	km/h	Delay
Koramangala → HSR Layout	**13.1**	10 min
Electronic City → HSR Layout	**13.1**	23 min
Majestic → MG Road	**13.2**	10 min
HSR Layout → Koramangala	**13.2**	10 min
MG Road → Majestic	**13.3**	10 min
Koramangala → Jayanagar	**13.4**	9 min

Of 90 corridors: 21 Critical (< 13.75 km/h), 46 Severe (< 15), 23 Moderate. Peak traffic factor reaches 2.6× at 19:00 on weekdays.

STAGE 3 — MODES

Bike vs auto vs cab

Bike

₹78
average fare

33% / 12%
of trips / of revenue

23 → 16.8
km/h, off-peak → peak

+47%
minutes per km in rain

Auto

₹172
average fare

35% / 28%
of trips / of revenue

19.5 → 12.8
km/h, off-peak → peak

+29%
minutes per km in rain

Cab

₹408
average fare

32% / 61%
of trips / of revenue

23.3 → 13.1
km/h, off-peak → peak

+31%
minutes per km in rain

STAGE 4

Machine learning models

MODEL A — REGRESSION

Trip duration (ETA)

Random Forest regressor · 80 trees, depth 12

- Features: vehicle, distance, hour, day, traffic factor, rain
- Learns actual ÷ formula ETA, so long trips still scale by vehicle
- Fare is then computed from the ETA via the rate card

2.995 min

Mean absolute error

0.957

R^2 score

MODEL B — CLASSIFICATION

Zone demand level → surge

Random Forest classifier · Low / Medium / High

- Trained on 38,158 MapReduce Job 4 cells
- Features: hour, day, weekend, peak, zone, rain
- Tertile labels: Low ≤ 4, Medium ≤ 8, High > 8 trips / hour

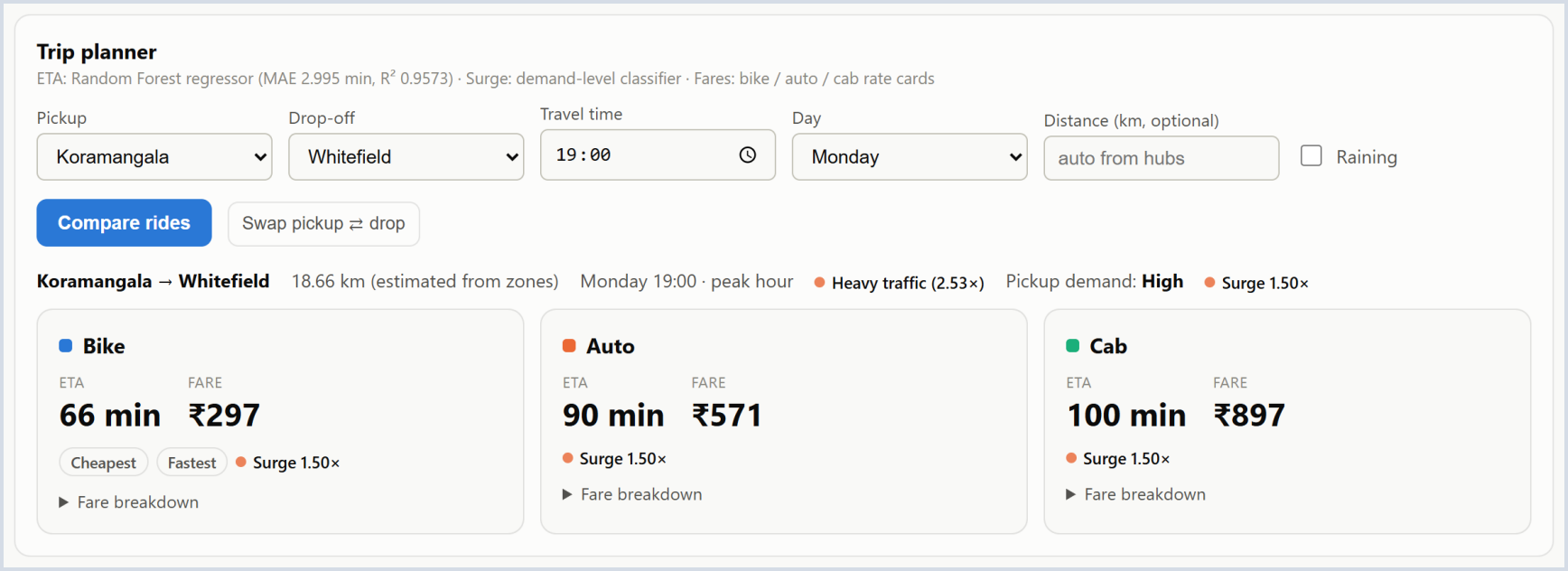
68.0%

Accuracy (random = 33%)

0.687

Weighted F1

Trip planner: compare all three rides



Koramangala → Whitefield, Monday 19:00: bike 66 min / ₹297, auto 90 min / ₹571, cab 100 min / ₹897. ETA from Model A, surge from Model B, fares from the rate card.

SO WHAT?

Insights & recommendations

1	**Put supply where the peaks are**	Shift drivers to 09:00–10:00 and 18:00–20:00; Job 1 shows these hours carry the most trips.
2	**Fix two corridor clusters first**	Koramangala / HSR / Electronic City (Silk Board area) and Majestic ↔ MG Road are the slowest, at 13.1–13.3 km/h.
3	**Recommend by context**	Bikes for short peak trips (16.8 km/h vs 13.1 for cabs); cabs in rain (bikes +47% min/km).
4	**Review peak-hour driver pay**	₹/trip-minute is lowest in jams; a higher delay rate would keep drivers on the road.
5	**Revisit the night auto surcharge**	At 1.5× an auto costs as much as a cab for 8 km+ trips after 22:00.

Thank you

GITHUB REPOSITORY

https://github.com/AnoonMR/bengaluru-urban-mobility-analytics